
Endocrine Pharmacology I – Disorders of the Thyroid Gland: Hyperthyroidism

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Learning goal

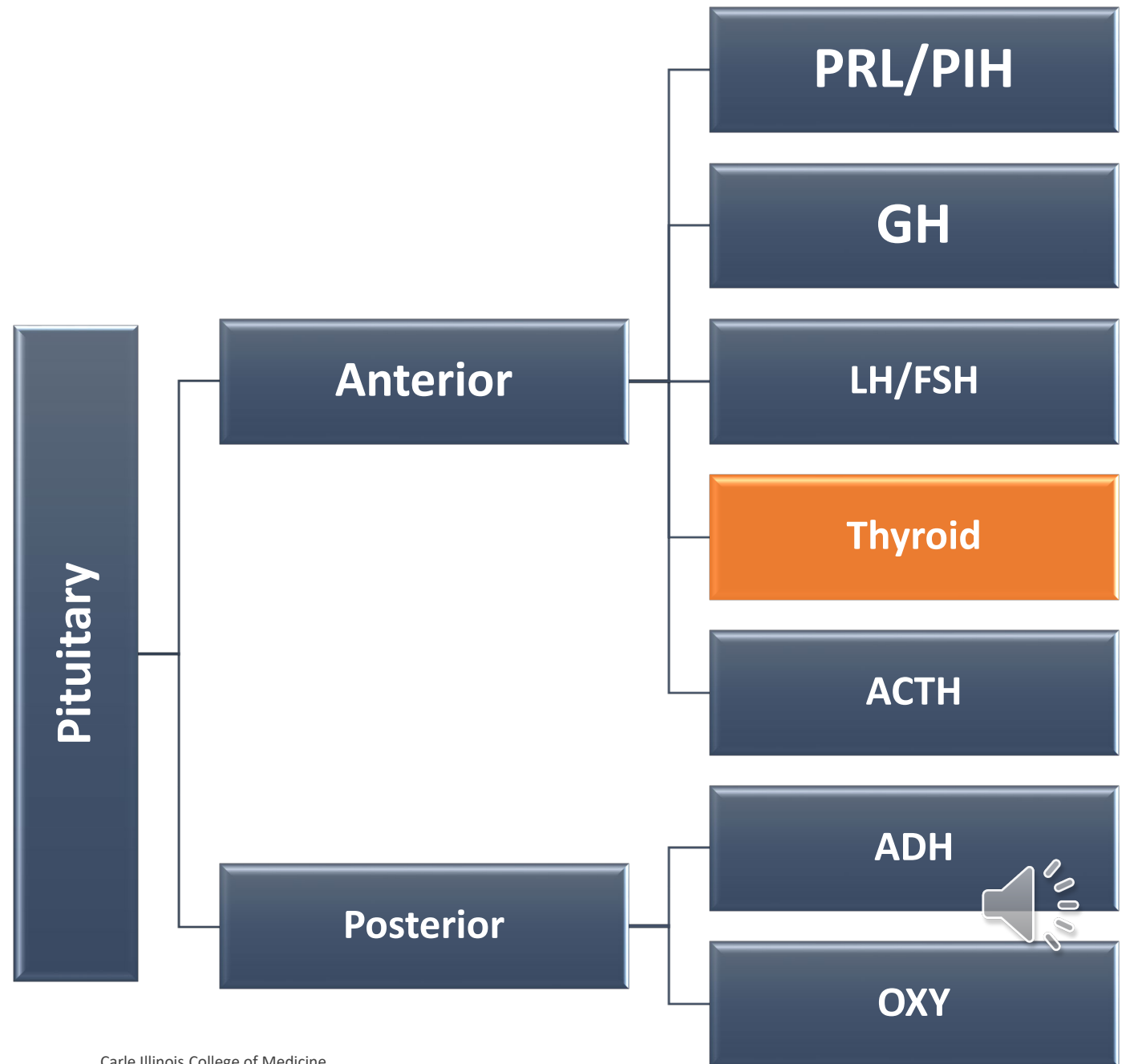
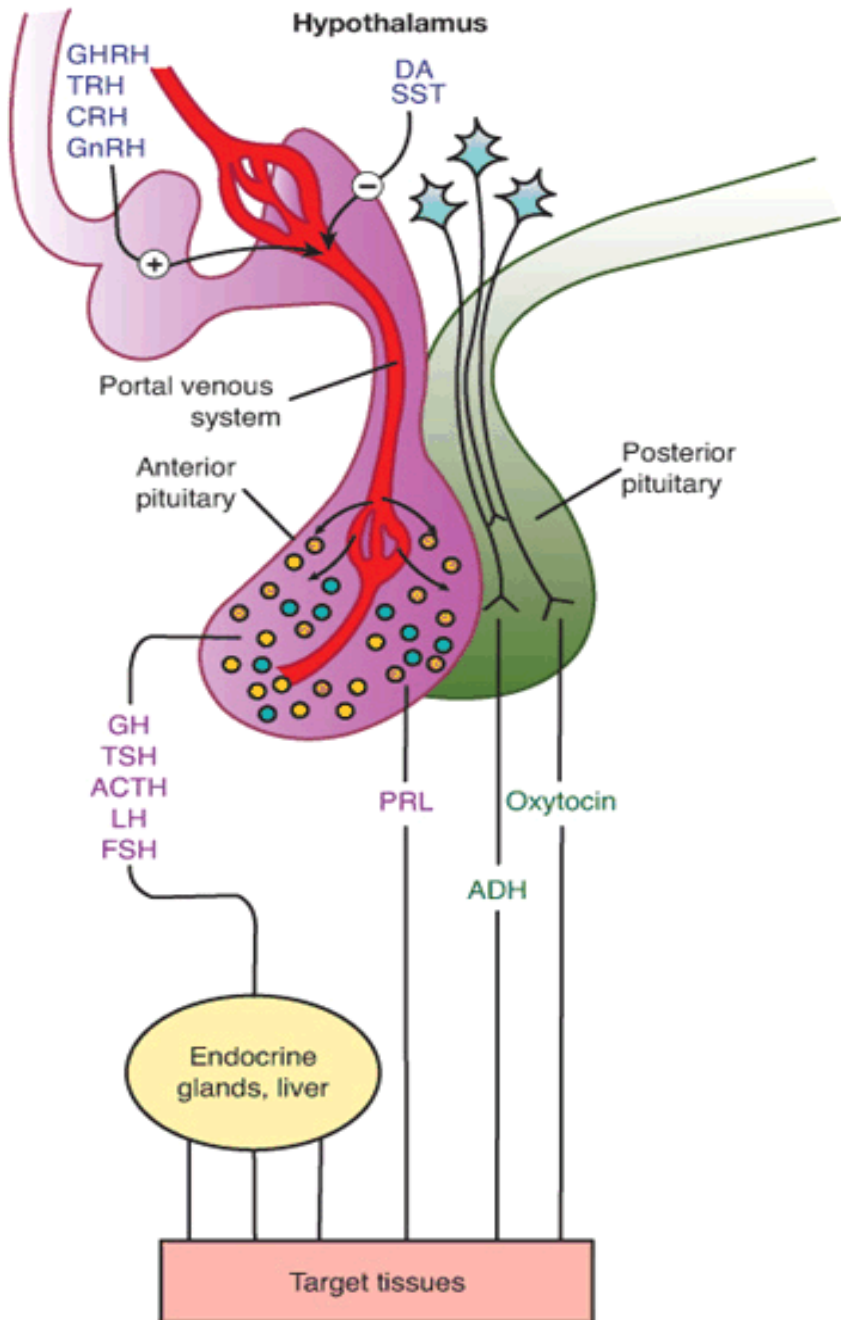
- **To rationally learn and recall the mechanisms of action, clinical indication, and adverse effects of drugs for the treatment of thyroid hormone excess.**



Learning objectives

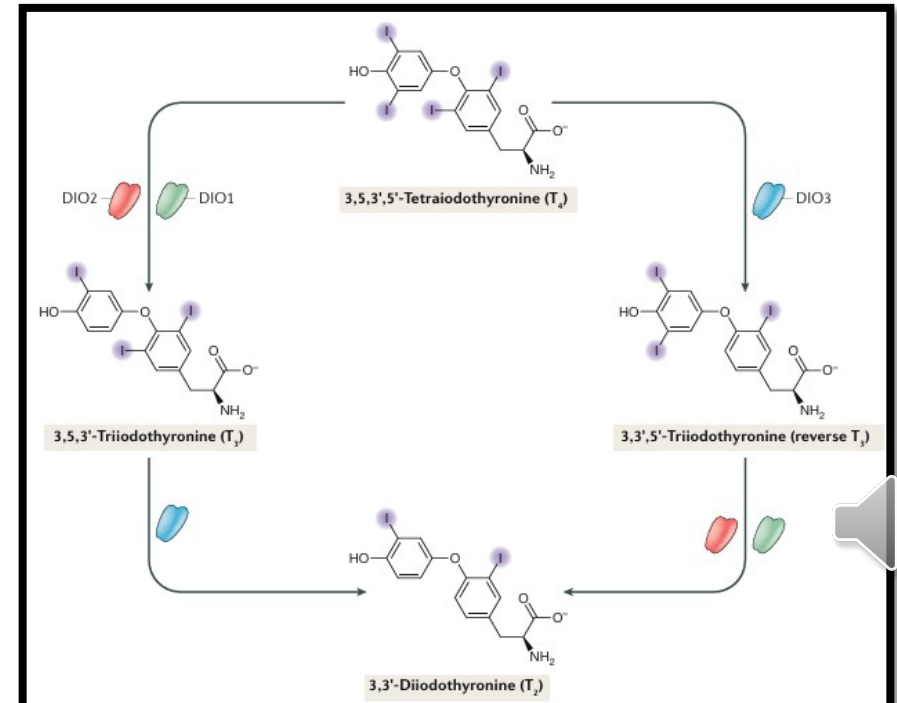
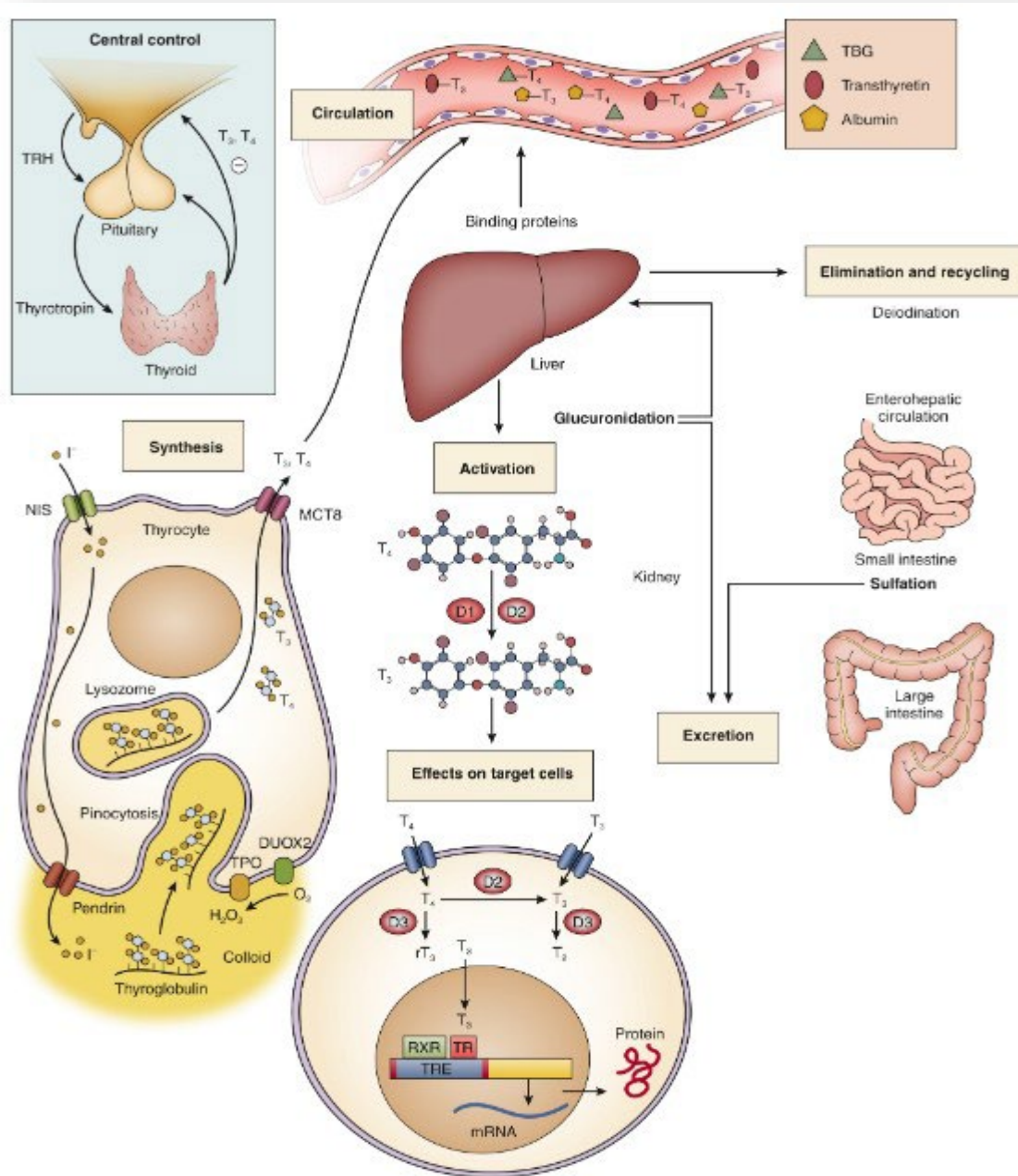
1. Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.
2. Critique the use of beta blockers in the management of thyrotoxicosis.

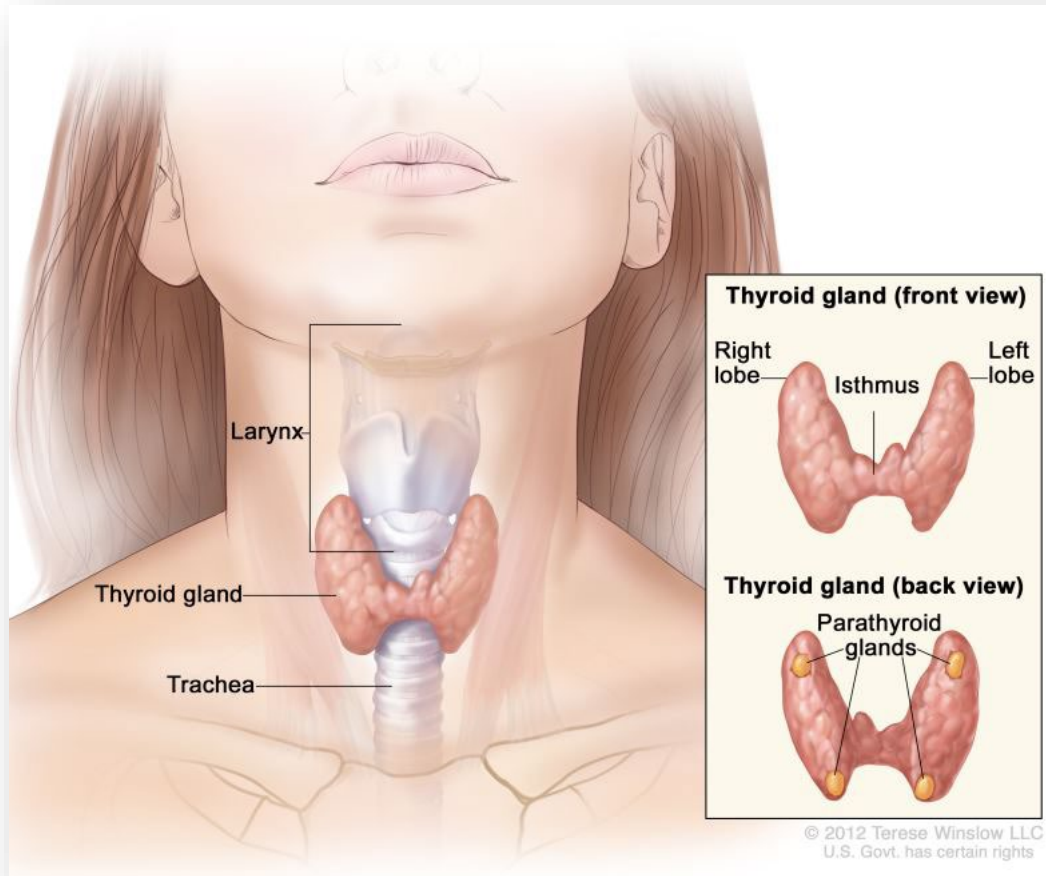




Review of thyroid hormone physiology

The physiologic actions of TH are produced predominately by T_3 with ~80% derived from T_4 deiodination in peripheral tissues



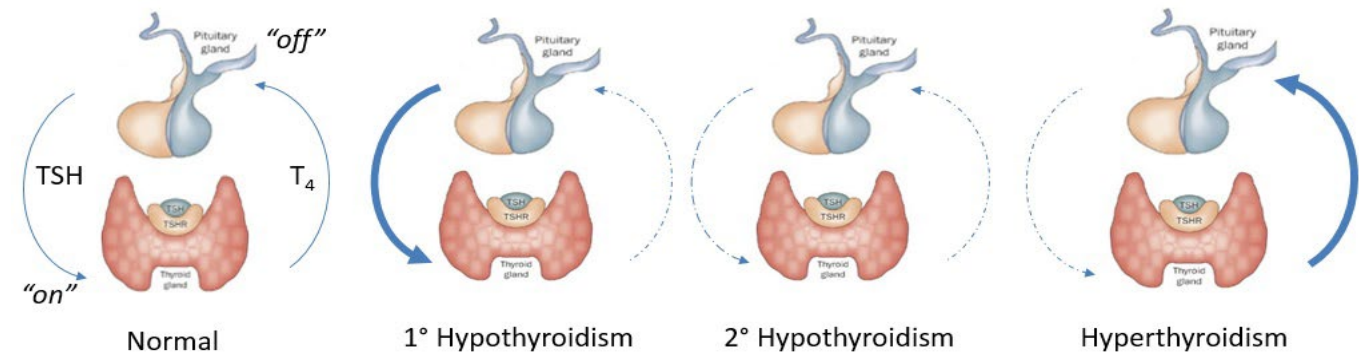


Thyroid Disease

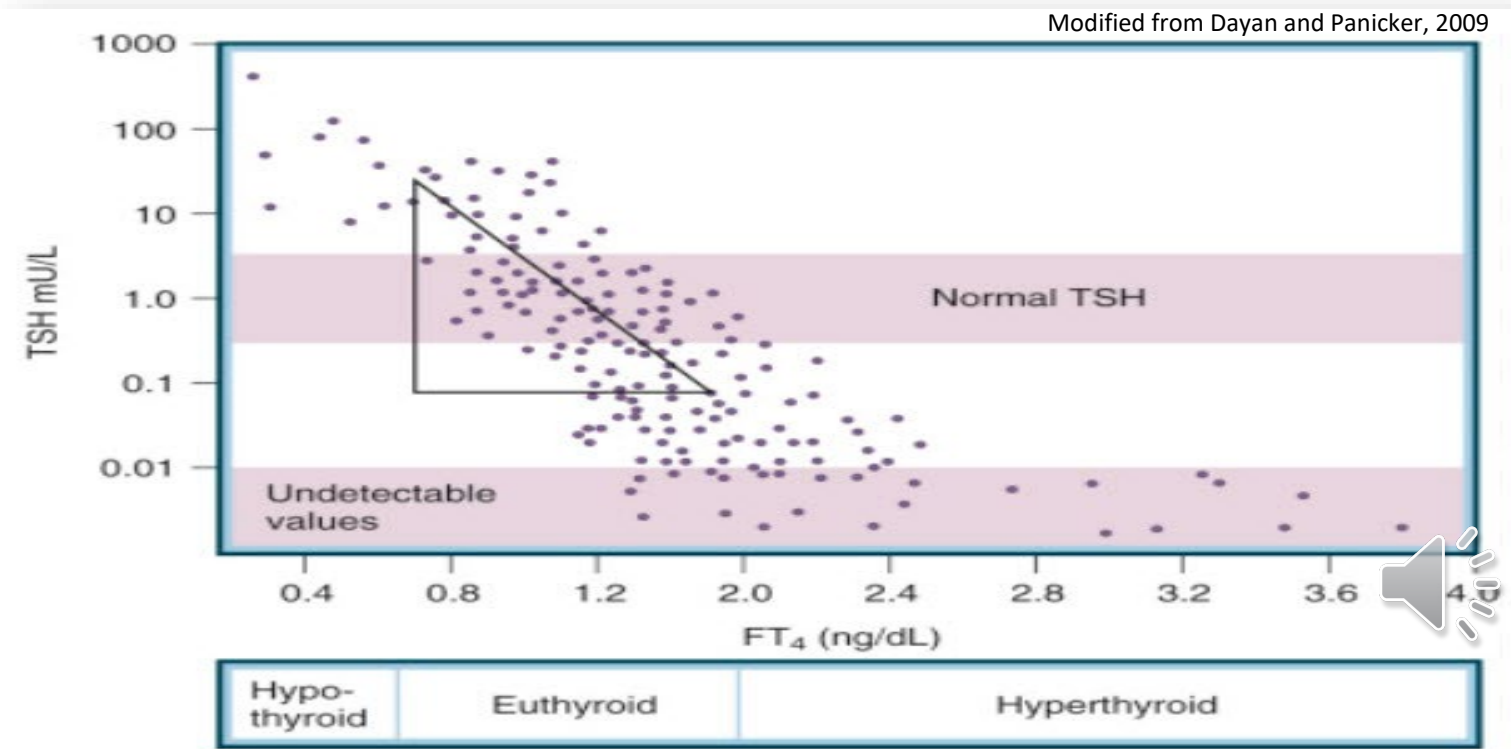
Therapeutics for thyroid hormone excess



The American Thyroid Association recommends initially checking **TSH levels** and **free-T₄** to test thyroid function.



TSH	Normal	High	Low	Low
T ₄	Normal	Low	Low	High



Thyrotoxicosis

Increased
thyroid
hormone
production

Release of
preformed
hormones
from thyroid
gland

Increases in
extrathyroidal
thyroid
hormone
availability

Surreptitious
thyroid
hormone
ingestion

Drug-induced

Struma ovarii



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

Drugs for thyrotoxicosis

Thyroid
hormone
synthesis
inhibitors

Radioactive
iodine

Inorganic
iodides

Beta-blockers

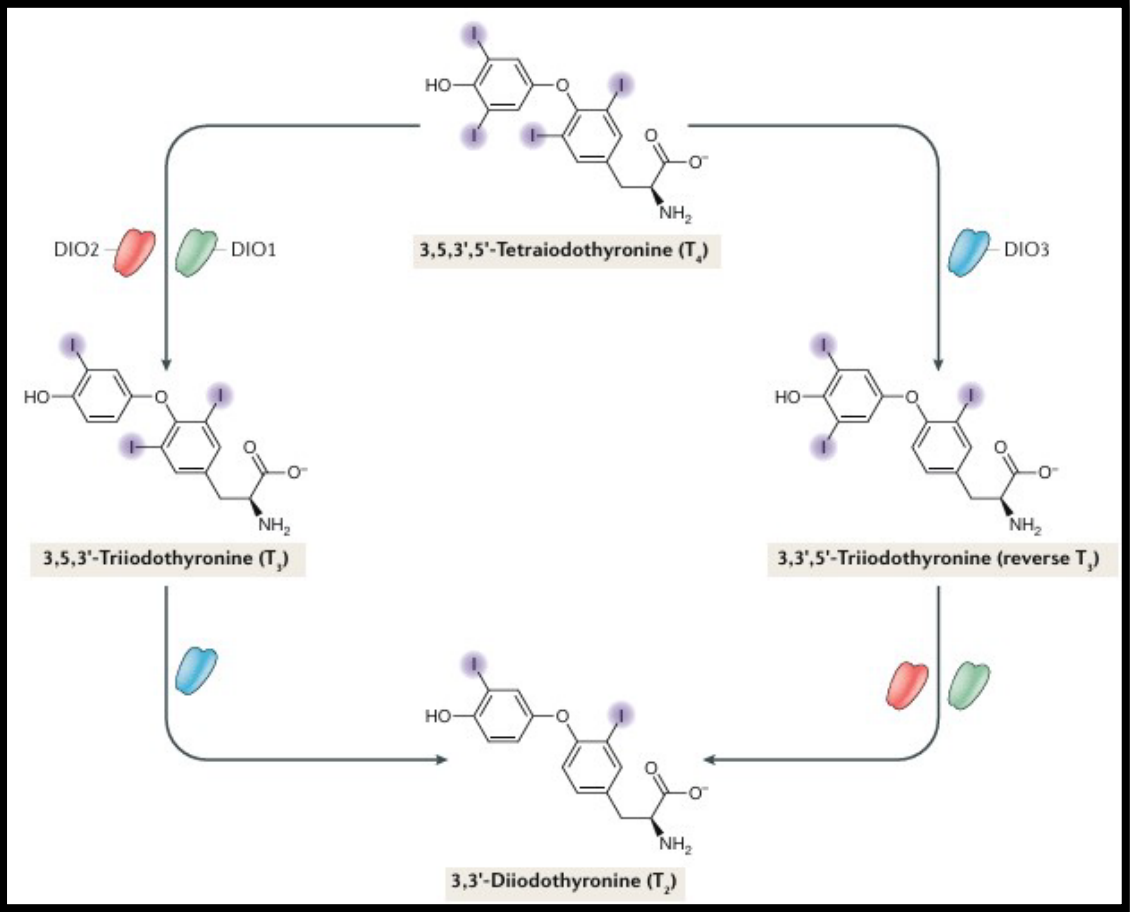
NSAIDs
Corticosteroids

***The goals of pharmacotherapy are to
eliminate the excess TH and minimize the symptoms and long-term
consequences of hyperthyroidism***

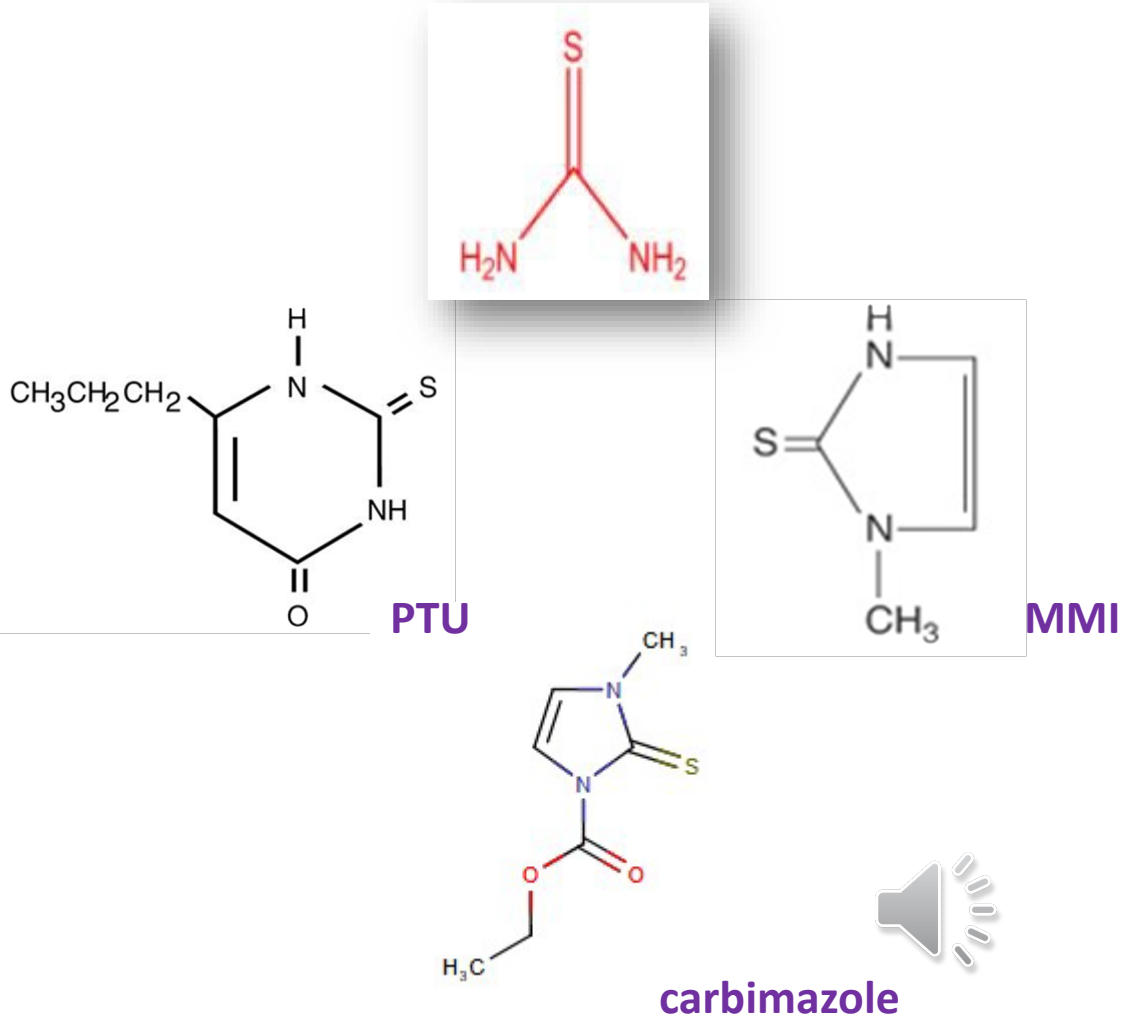


Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

“Thionamides”



Luongo et al. 2019



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

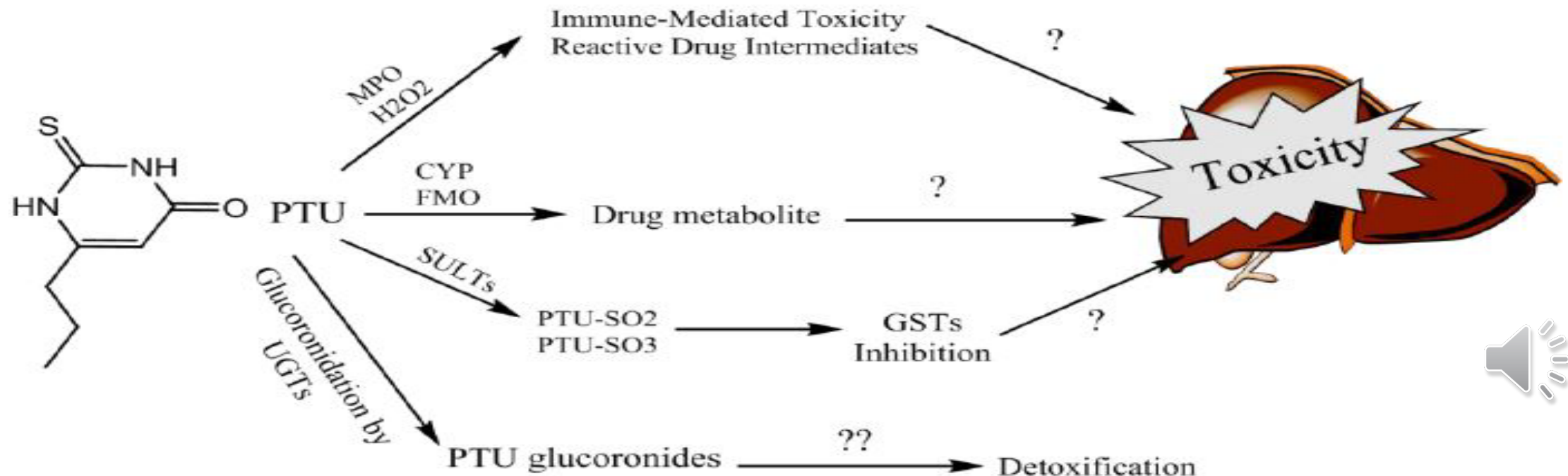
	Propylthiouracil (PTU)	Methimazole (MMI)
Inhibition of type 1 deiodinase	Yes	No
Route of administration	Oral	Oral
Relative Potency	1X	>10X (up to 50X)
Absorption	Nearly complete	Nearly complete
Protein binding	75%	Negligible
Onset	10 – 21 days	1 week
Serum half-life (h)	1 – 2	4 – 6
Duration of action (h)	12 – 24	>24
Cross the placenta?	Yes, at low levels	Yes, at low levels
Breastmilk levels	Lower	Low
Compliance	Worse (3 or 4X daily)	Better (1 or 2X daily)
Cost	Less expensive	More expensive
¹³¹I outcome	Decreases effect	No effect



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

WARNING

Severe liver injury and acute liver failure, in some cases fatal, have been reported in patients treated with propylthiouracil. These reports of hepatic reactions include cases requiring liver transplantation in adult and pediatric patients. Propylthiouracil should be reserved for patients who cannot tolerate methimazole in in who radioactive iodine therapy or surgery are not appropriate treatments for the management of hyperthyroidism. Propylthiouracil may be the treatment of choice when an antithyroid drug is indicated during or just prior to the first trimester of pregnancy.



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

MMI and PTU are associated with common adverse effects such as rash and pruritus.

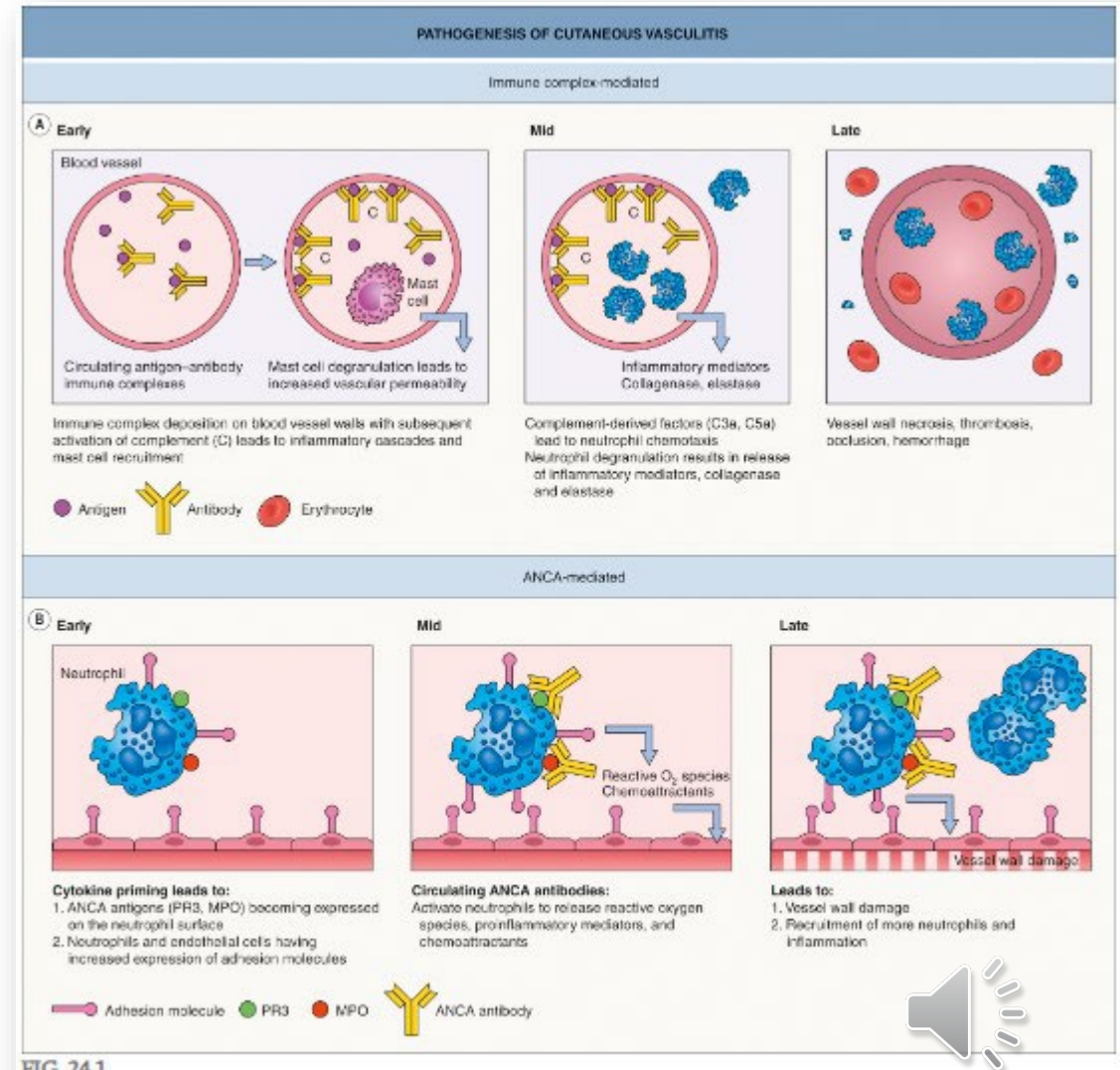


FIG. 24.1

Pathogenesis of cutaneous vasculitis— immune complex- versus anti-neutrophil cytoplasmic antibody (ANCA)-mediated .

Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

Radioactive iodine (^{131}I) (sodium iodide 131)

- Most commonly used therapy
- Used in those intolerant to antithyroid drugs
- 90+% efficacy
- Results in thyroid tissue destruction
- Normalization of thyroid function within 6 – 12 months

Can cause or exacerbate Graves eye disease

Permanent hypothyroidism in treatment for Graves disease

Risk of breast cancer and other solid tumor development

ABSOLUTELY CONTRAINDICATED IN PREGNANCY

Inorganic iodide solutions

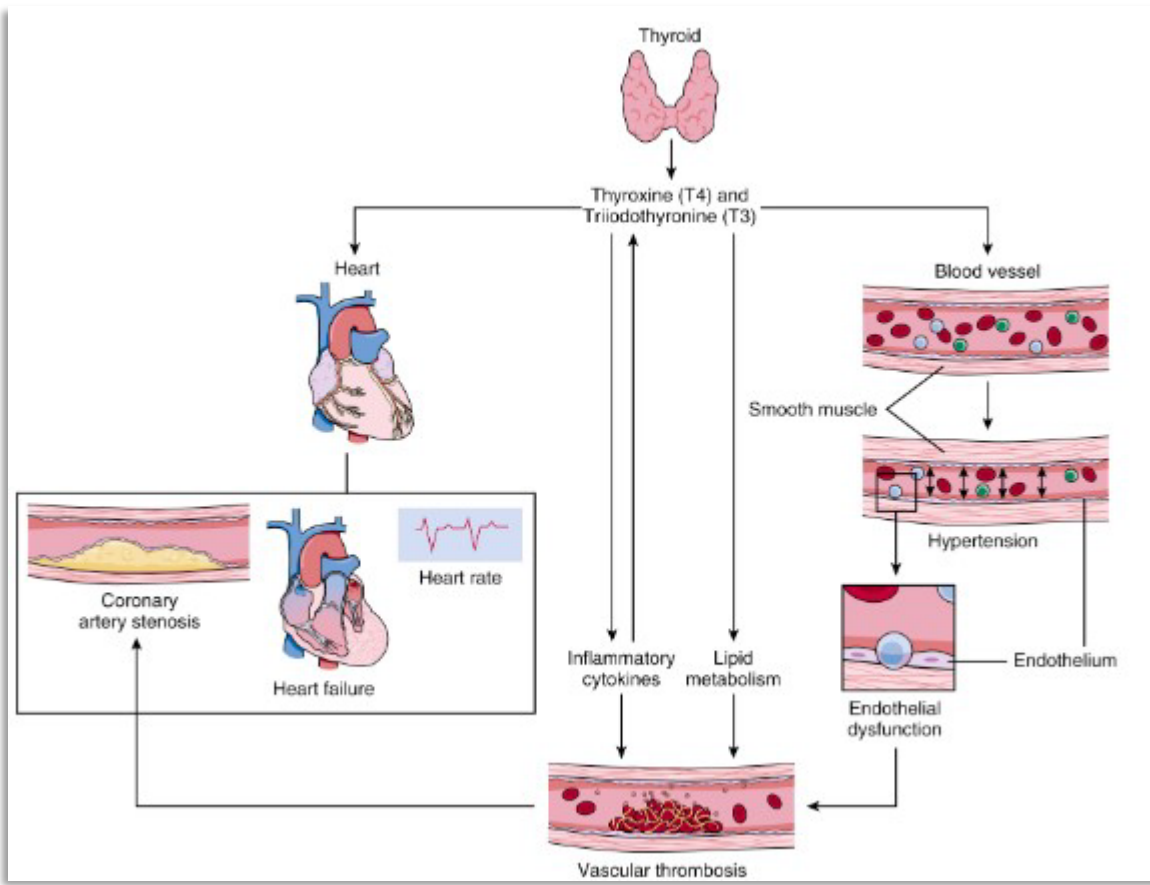
- Inhibit the release of stored thyroid hormone
- Lugol's solution
 - Iodine, KI, and water
- Saturated solution of KI (SSKI)
- Reserved for severe thyrotoxicosis and surgery in Graves disease management



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

Beta-adrenergic receptor blockers

- Control cardiac and psychomotor manifestations of thyrotoxicosis within minutes
- Inhibit 5'-deiodinase to decrease peripheral conversion of T_4 to T_3
- **Propranolol**, atenolol, metoprolol, nadolol, esmolol
- Can cause bradycardia, hypotension, AV block, bronchospasm, CNS adverse effects, diarrhea, and nausea.



NSAIDs and Steroids

For the moderate-to-severe pain with symptomatic thyrotoxicosis

NSAIDs

Corticosteroids

- Inhibit 5'-deiodinase to decrease peripheral conversion of T_4 to T_3
- **Short-term** use of prednisone or hydrocortisone
- Caution with pregnancy



Differentiate between the pharmacologic treatment modalities for hyperthyroidism including the drugs of choice and their clinical pharmacology (pharmacokinetics and pharmacodynamics) in non-pregnant patients.

Thyroid storm

Life-threatening emergency



Patients present with CNS, GI, and CV symptoms

Precipitated by:

Surgery
Amiodarone
Discontinuation of thionamides

Thionamides*

Iodides
(1 hr post thioamides)

Corticosteroids

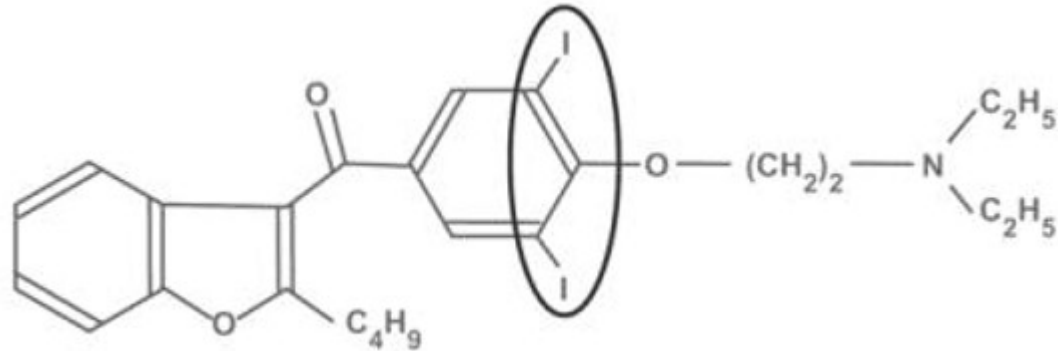
Beta-blocker

Cholestyramine
Acetaminophen

* PTU preferred

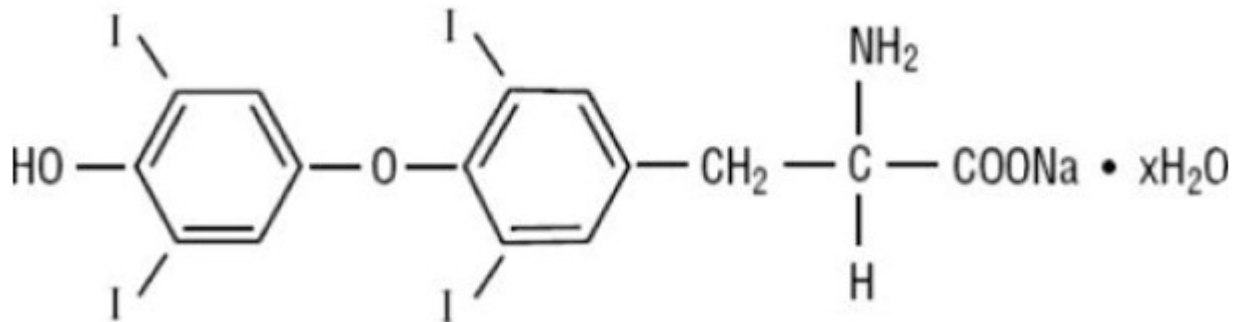
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Amiodarone-induced thyrotoxicosis (AIT)



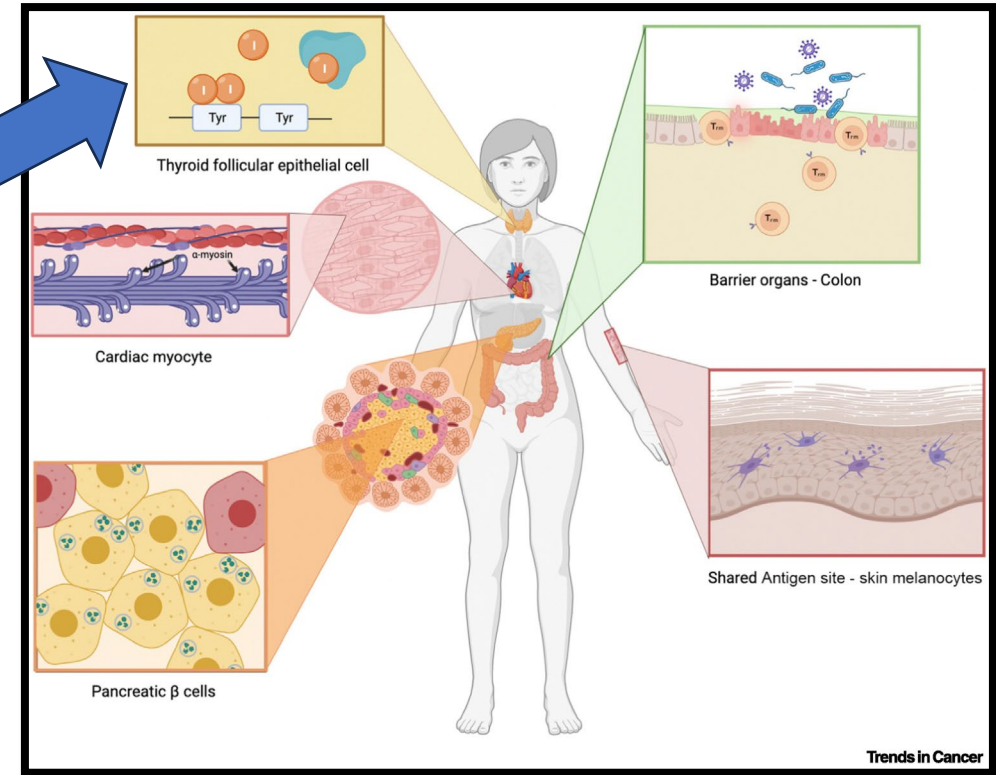
- **An antiarrhythmic**
- **Structure analogous to T₄**
- **High iodine content**
- **Lipophilic**
- **Long half-life**
- **Accumulates in various organs**
- **Type I and Type II AIT**

Levothyroxine (T₄) Sodium



Immune checkpoint inhibitors (ICI) and thyroid dysfunction

Hyperthyroidism Thyrotoxic phase of thyroiditis



Most patients may progress to develop primary hypothyroidism but a few may need to be evaluated for the development of Graves disease

Thyrotoxicosis

Graves
disease

Toxic
multinodular
goiter

Solitary toxic
nodules

Thyroiditis

Amiodarone-
induced
thyrotoxicosis

Immune
checkpoint
inhibitor-
induced

antithyroid drugs, radioactive iodine, or surgery

Type I: MMI
Type II: Steroids

Watchful *waiting*



Key points/summary

- The most common cause of hyperthyroidism is Graves disease.
- Pharmacologic treatment of hyperthyroidism should be individualized.
- Hyperthyroidism can be treated with PTU or MMI
 - They inhibit the synthesis of TH by preventing the incorporation of I into iodotyrosines and by inhibiting the coupling of monoiodotyrosine and diiodotyrosine to form T_4 and T_3
 - Unlike PTU, MMI does not inhibit the peripheral deiodination of thyroxine to triiodothyronine.
- Propylthiouracil has a boxed warning of severe liver injury and acute liver failure, with some fatalities reported.
- The presentation of thyroid storm typically involves thyrotoxicosis as an underlying condition, along with a precipitating agent, such as surgery or the use of amiodarone
 - Hyperpyrexia and cardiovascular and CNS dysfunction occurs

